Prithvi Sunilkumar

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Education

Carnegie Mellon University **Pittsburgh, PA**
Master of Information Systems Management | 25% Merit Scholarship *August 2026 – December 2027*

- Courses: Generative AI Lab, Introduction to Machine Learning, Intermediate Database Management, Time Series Forecasting, Responsible AI

PES University **Bengaluru, India**
Bachelor of Technology in Computer Science and Engineering | GPA: 3.56/4.00 *August 2019 – May 2023*

- Courses: Python, Data Analytics, Database Management Systems, Design and Analysis of Algorithms

Professional Experience

Rakuten **Bengaluru, India**
Software Engineer *July 2023 – July 2026*

- Built **QCC Workflow Automation** from **0** $\rightarrow$ **1**, an approval-workflow platform adopted across **Rakuten Group** divisions worldwide; re-architected its AI layer from a single text-completion call into a **RAG pipeline** over years of historical consultant feedback, cutting manual review time **25%**
- Engineered the **Angular** frontend and **Node.js** services for QCC, integrating **OpenAI** models through Rakuten's internal API gateway to assist consultants in drafting grounded, consistent feedback
- Architected **OKR Dashboard 3.0** from the ground up — **React** frontend, **REST APIs**, and database layer — for Rakuten Group's company-wide goal-tracking platform, used by teams across **Japan, India, and APAC**
- Cached query results and sessions in **Redis** to cut repeat database reads, instrumented the platform with **OpenSearch** and **Grafana** for log monitoring, and built the interface directly from **Figma** designs via Figma's **MCP integration**
- Delivered **20+ proof-of-concept deployments** for external enterprise customers on **Rakuten SixthSense**, a commercial full-stack **observability** platform, across Application Observability and Log Monitoring
- Constructed custom **OpenTelemetry** telemetry components resolving client-specific instrumentation gaps that native agents did not cover, accelerating early product adoption
- Automated build and release with **Jenkins** CI/CD pipelines running **automated tests** and deploying **Dockerized** microservices to on-premise staging and production on merge, replacing a fully manual release process

Software Engineer Intern *January 2023 – June 2023*

- Launched the first version of QCC Workflow Automation, replacing manual consultant approval workflows
- Implemented UI components and REST APIs using **React** and **Node.js**; converted to a full-time offer

Diageo **Bengaluru, India**
Business Analyst Intern *June 2022 – August 2022*

- Designed an IT Service Management **Power BI** dashboard on **ServiceNow** data, surfacing ticket volumes, resolution times, and SLA compliance, and presented recommendations to IT stakeholders

Projects

Accessibility Bug Triage Harness | **Easy Alliance** | *Python, LLM APIs, WCAG 2.2, statistical evaluation* **2026 – Present**

- Building an **LLM triage system** that classifies free-text bug reports by whether they describe a real barrier, which **WCAG 2.2** success criterion they fail, severity, and reproducibility
- Constraining the model to narrow **typed questions** with all routing logic in deterministic Python, using **two-stage WCAG mapping** (guideline, then criterion) to improve accuracy over direct prediction
- Measuring whether model confidence is reliable enough to route on, reporting accuracy, **expected calibration error**, **Brier score**, and coverage-versus-accuracy across a hash-partitioned dev and held-out test split
- Hardening evaluation integrity and cost controls: a **leakage filter** dropping reports that already cite a criterion, result caching, pre-flight cost estimates, and spend caps, benchmarked against a **Claude Haiku** baseline

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS
Frontend & Backend: React, Angular, Node.js, Express.js, REST APIs, Redis
AI & Data: RAG pipelines, LLM API integration, LLM evaluation (calibration, ECE, Brier), prompt design, Power BI
Cloud & Observability: Docker, Jenkins CI/CD, AWS, OpenTelemetry, OpenSearch, Grafana
Tools & Practices: Git, Jira, Agile/Scrum, Figma, MCP integrations, Linux